

LAB_08 : Calculate use Python in Linux environment

Use your Linux environment and vi editor for the following:

1. Write a program using a loop statement to calculate the average of the following numbers: 9, 41, 12, 3, 74, 15. You can use these numbers as a set in the for loop.

Solution :

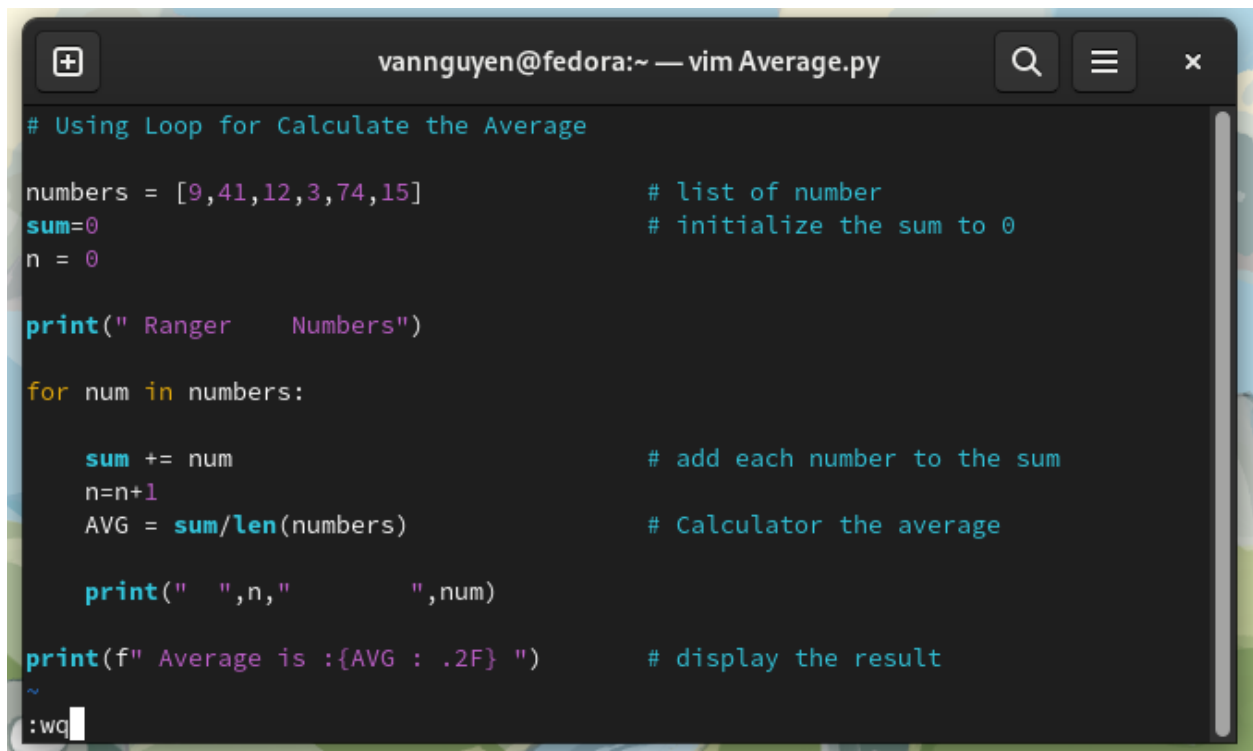
```
$ vim Average.py          : Create a file Vi editor ".py"
                           ( i : inset )
```

```
numbers = [9,41,12,3,74,15]
sum = 0
for num in numbers:
    sum += num
    AVG = sum/len(numbers)
Print(" Average is : ",AVG)
```

```
$ python3 Average.py      : Run file
```



Input



```
vannguyen@fedora:~ — vim Average.py

# Using Loop for Calculate the Average

numbers = [9,41,12,3,74,15]          # list of number
sum=0                                # initialize the sum to 0
n = 0

print(" Ranger      Numbers")

for num in numbers:

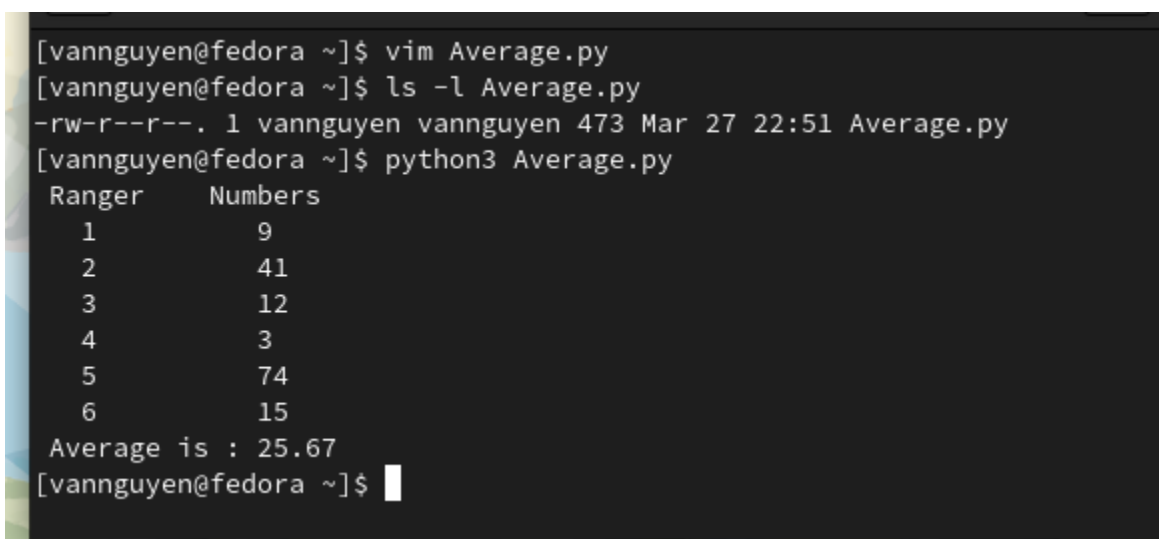
    sum += num                        # add each number to the sum
    n=n+1
    AVG = sum/len(numbers)           # Calculator the average

    print("   ",n,"          ",num)

print(f" Average is :{AVG : .2F} ")   # display the result

~
:wq
```

Output



```
[vannguyen@fedora ~]$ vim Average.py
[vannguyen@fedora ~]$ ls -l Average.py
-rw-r--r--. 1 vannguyen vannguyen 473 Mar 27 22:51 Average.py
[vannguyen@fedora ~]$ python3 Average.py
Ranger      Numbers
 1           9
 2          41
 3          12
 4           3
 5          74
 6          15
Average is : 25.67
[vannguyen@fedora ~]$
```



2. Write a function to pass three numbers (int or float) as arguments and calculate their average. You can select the function name. The three numbers should be given as input from the user.

Solution :

```
$ vim AVG_3_numbers.py          : Create a file Vi editor ".py"
                                ( i : inset )
```

```
def calculate_avg(num1, num2, num3):
    sum = num1 + num2 + num3
    average = sum/3
    return average

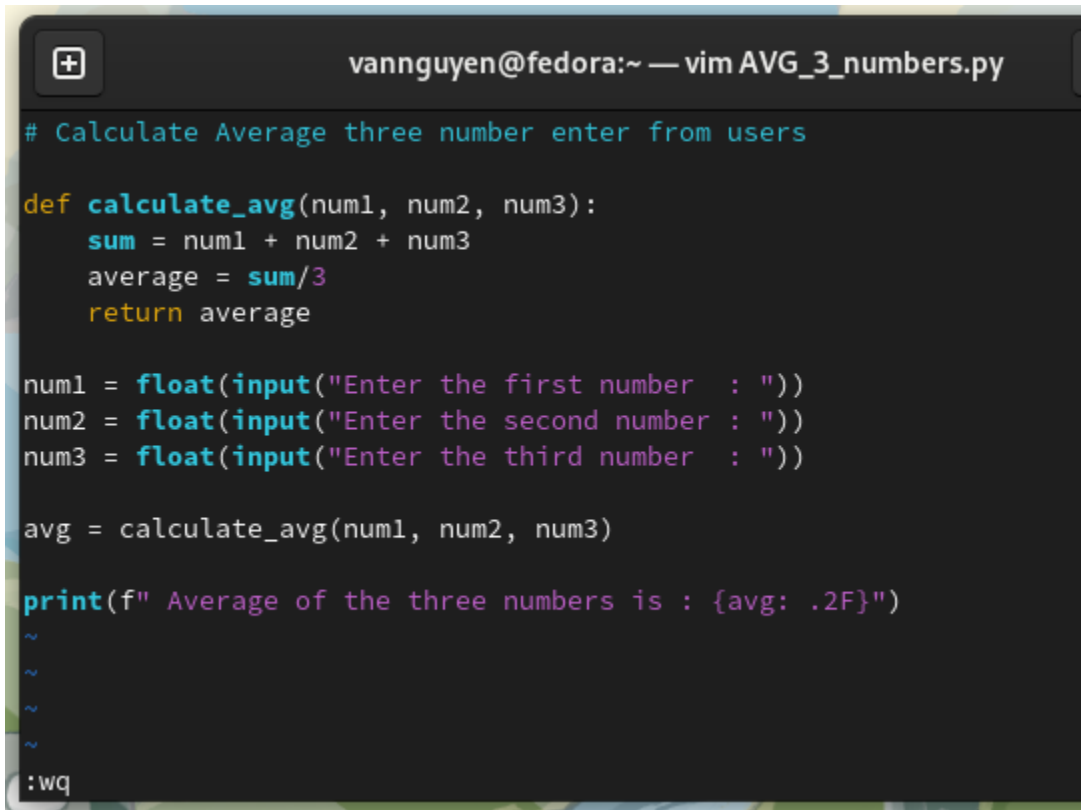
num1 = float(input(" Enter the first number :" ))
num2 = float(input(" Enter the second number:" ))
num3 = float(input(" Enter the third number :" ))

avg = calculate_avg(num1, num2, num3)

Print(" Average of three numbers is : ",avg)
```

```
$ python3 AVG_3_numbers.py      : Run file
```



Input :

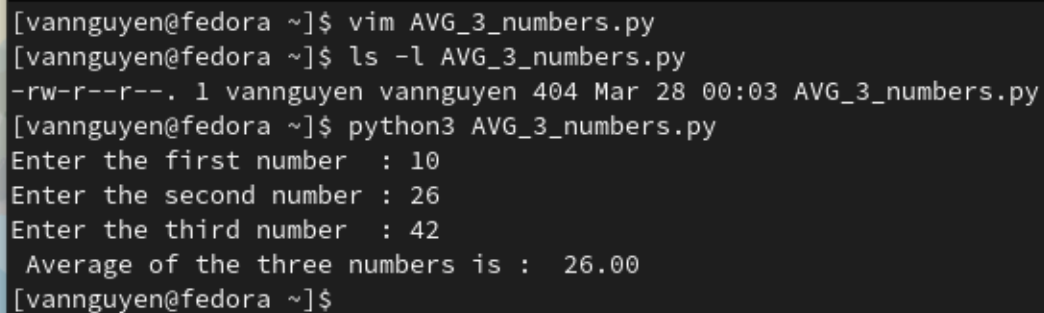
```
vannguyen@fedora:~ — vim AVG_3_numbers.py
# Calculate Average three number enter from users

def calculate_avg(num1, num2, num3):
    sum = num1 + num2 + num3
    average = sum/3
    return average

num1 = float(input("Enter the first number : "))
num2 = float(input("Enter the second number : "))
num3 = float(input("Enter the third number : "))

avg = calculate_avg(num1, num2, num3)

print(f" Average of the three numbers is : {avg: .2F}")
~
~
~
~
:wq
```

Output :

```
[vannguyen@fedora ~]$ vim AVG_3_numbers.py
[vannguyen@fedora ~]$ ls -l AVG_3_numbers.py
-rw-r--r--. 1 vannguyen vannguyen 404 Mar 28 00:03 AVG_3_numbers.py
[vannguyen@fedora ~]$ python3 AVG_3_numbers.py
Enter the first number : 10
Enter the second number : 26
Enter the third number : 42
Average of the three numbers is : 26.00
[vannguyen@fedora ~]$
```

